AUTOMATIC PILL DISPENSER

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PROBLEM STATEMENT

In this modern era humans are largely dependent on the use of medicines. Studies show that many people irrespective of age become forgetful when it comes to consuming medicines. This is seen mostly in the elderly, leads to many other life risks.

INTRODUCTION

This project presents an IoT-based automatic pill dispenser to ensure timely medication intake. The system dispenses medicines at scheduled times using a real-time clock. It also provides alerts and notifications through a buzzer, display, and Telegram chatbot, reducing missed doses.

OBJECTIVES

- **To develop an automatic tablet dispensing system using an RTC-based control mechanism for accurate medicine dispensing at scheduled times.**
- **To provide storage and dispensing of tablets for up to 3 days with 3 doses per day in an organized manner.**
- **To implement a user notification system using a display and buzzer to indicate dispensing status, alerts, and other system information using Telegram Bot.**

LITERATURE SURVEY

TITLE OF PAPER	RESEARCH CONTRIBUTIONS	RESEARCH GAP
Design and Implementation of an Automatic Pill Dispenser for Elderly Care (IEEE, 2018)	• Automated pill dispensing • Audio visual medication alert • Reduces human error	• No remote monitoring • No emergency alert system
IoT-Based Smart Medicine Dispenser System (IEEE, 2019)	• Mobile app integration • Real time reminders • Improved adherance	• High system cost • Internet dependency

LITERATURE SURVEY

Arduino-Based Automatic Medication Dispenser with Reminder System (IEEE, 2017)	• Time-based automatic pill dispensing • LCD-based dosage display and buzzer alerts • Simple and low-cost implementation	• No IoT connectivity • No remote monitoring or alert system
Smart IoT Medicine Dispenser with Monitoring and Alert System (IEEE, 2020)	• IoT-based remote monitoring • Real-time alerts and notifications • Integration with mobile/web interface	• Complex system design • Higher cost and power consumption

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS			
SI NO.	**SYSTEM FUCTION**	**REQUIREMENT**	**COMPONENT USED**
1.	The system requires a central controller to manage all operations including timing, motor control, user interaction, and communication. .	A microcontroller with sufficient GPIO pins,acess to wifi, processing capability, and communication support.	ESP32 DevKit V1
2.	The system needs accurate time tracking to ensure medicines are dispensed at the correct scheduled time without manual intervention.	A real-time clock module with high accuracy and battery backup.	DS3231 RTC Module
3.	The system must rotate the pill storage container precisely to align the correct compartment for dispensing at the scheduled time.	A stepper motor capable of controlled step-by-step rotation.	28BYJ-48 Stepper Motor

SI NO.	SYSTEM FUNCTION	REQUIREMENT	COMPONENT USED
4.	The system requires a mechanism to open or release pills once the correct compartment is positioned. This ensures controlled dispensing.	A servo motor capable of precise angular movement.	SG90 Servo Motor
5.	The system needs an interface to drive the stepper motor safely, as the controller cannot supply required current directly.	A motor driver module to amplify control signals and drive the motor.	ULN2003 Motor Driver
6.	The system should provide real-time information such as time, alerts, and system status to the user in a clear and readable format	A compact, low-power display module with good visibility.	SSD1306 OLED Display
7.	The system must alert the user when it is time to take medicine and allow confirmation after intake to ensure compliance tracking.	An audible buzzer for alerts and a push button for user confirmation input.	Buzzer + Push Button

SOFTWARE REQUIREMENTS

SI NO.	SYSTEM FUNCTION	REQUIREMENT	COMPONENT USED
1.	The system needs a remote notification mechanism to alert users about medication timings even when they are not near the device.	Integration with an online messaging platform for sending alerts	Telegram Bot
2.	The system requires a development environment to write, compile, debug, and upload the firmware to the microcontroller	An IDE that supports ESP32 and required libraries	Arduino IDE 2.3.8
3.	The system logic must handle scheduling, motor control, alert generation, and user interaction efficiently in real time.	A programming language suitable for embedded systems and hardware control.	C

FORMULA AND CALCULATIONS

Angle per step calculation:

$$\text{Step Angle} = 360/2048 = \sim 0.175°$$

This is based on the total number of steps required for one full rotation of the stepper motor

Angle for required steps:

$$\text{Rotation Angle} = \text{Number of Steps} \times \text{Step Angle}$$

$$= 227 \times 0.175\circ = 39.7\circ \approx 40°$$

Torque Estimation

$$T=F*r$$

where,T=Torque,F=Force acting ,r=Radius of the container

The force acting on the container is given by

$$F =mg$$

The mass of the container is **0.25 kg**

$$F =0.25×9.81$$

$$F =2.45N$$

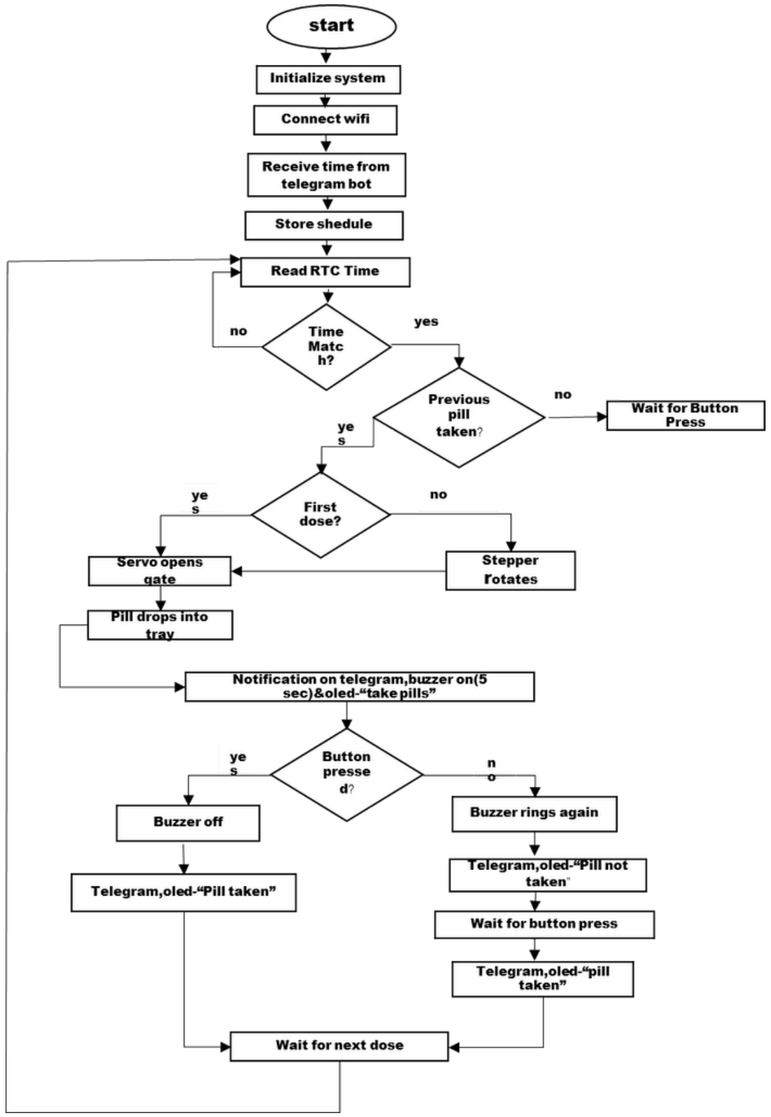
The container radius is **0.1 m**

$$T =2.45×0.1$$

$$T \approx 0.245Nm$$

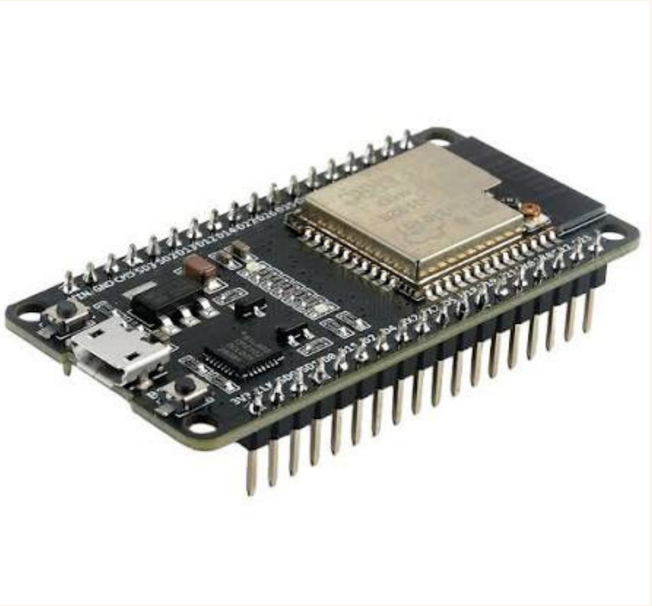
Thus the selected stepper motor provides sufficient torque to rotate the pill container.

FLOW CHART


start
Initialize system
Connect wifi
Receive time from telegram bot
Store shedule
Read RTC Time
Time Matc h?
no
yes
Previous pill taken?
no
Wait for Button Press
ye s
First dose?
ye s
no
Servo opens gate
Stepper rotates
Pill drops into tray
Notification on telegram,buzzer on(5 sec)&oled-"take pills"
Button presse d?
ye s
n o
Buzzer off
Buzzer rings again
Telegram,oled-"Pill taken"
Telegram,oled-"Pill not taken"
Wait for button press
Telegram,oled-"pill taken"
Wait for next dose

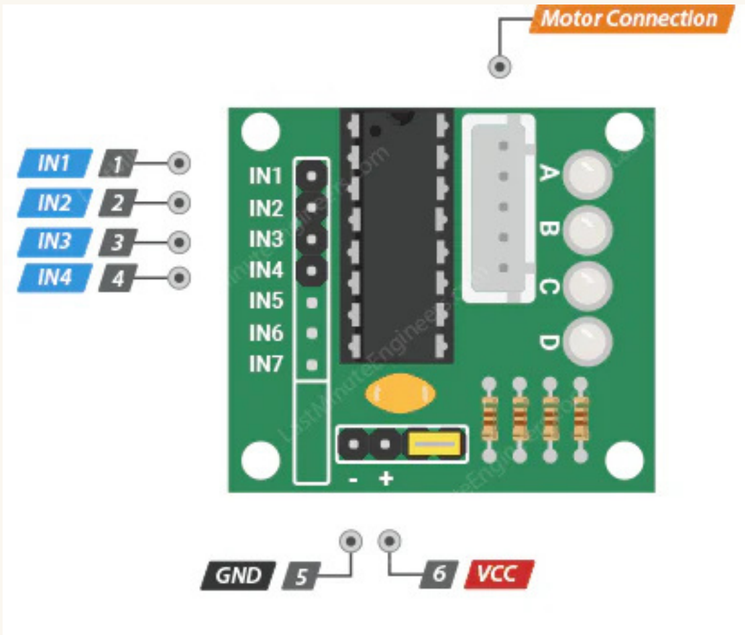
HARDWARE SPECIFICATIONS

ESP32 Dev Module



- MCU: Xtensa dual-core 32-bit LX6, up to 240MHz
- ADC: 12-bit, 18 channels, ADC1 (GPIO32–39) used — isolated from WiFi interference
- ADC Range: 0–3.3V with 11dB attenuation.
- ADC Resolution: 4096 levels (12-bit)
- Supply: 3.3V (sensors) + 5V USB (board)
- I/O Voltage: 3.3V logic
- Used Pins: GPIO32 (current), GPIO35 (voltage), GPIO21/22 (OLED I2C)
- Key Feature: 64× oversampling used in software to improve effective ADC
- resolution.

Motor driver - ULN2003



- Type: Darlington transistor array
- Number of channels: 7
- Input voltage: 3.3V / 5V compatible
- Output current: up to 500 mA per channel
- Output voltage: up to 50V
- Function: Drives stepper motor by amplifying current
- Protection: Internal flyback diodes

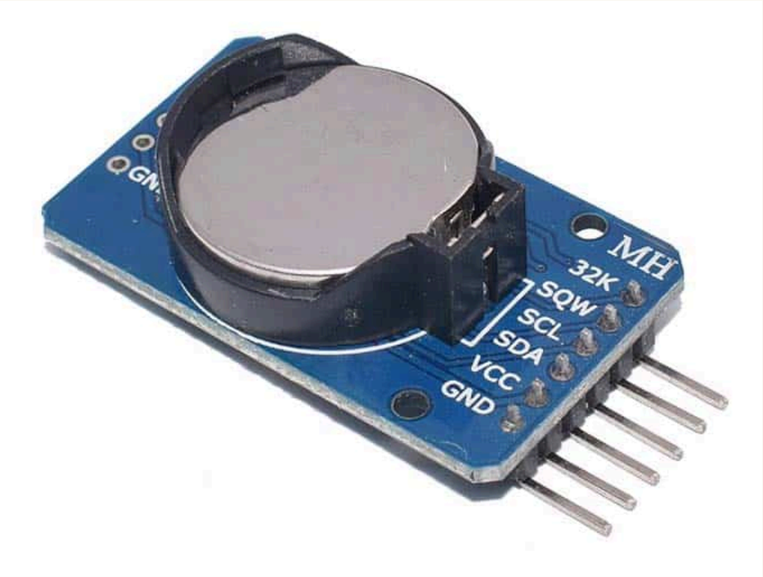
Stepper motor

- Type: Unipolar stepper motor
- Operating voltage: 5V DC
- Step angle: 5.625° per step
- Gear reduction ratio: 1:64
- Effective steps per revolution: 2048 steps
- Current consumption: ~240 mA
- Speed: ~10–15 RPM
- Control: Requires driver (ULN2003)
- Rotation: Step-wise (precise control)



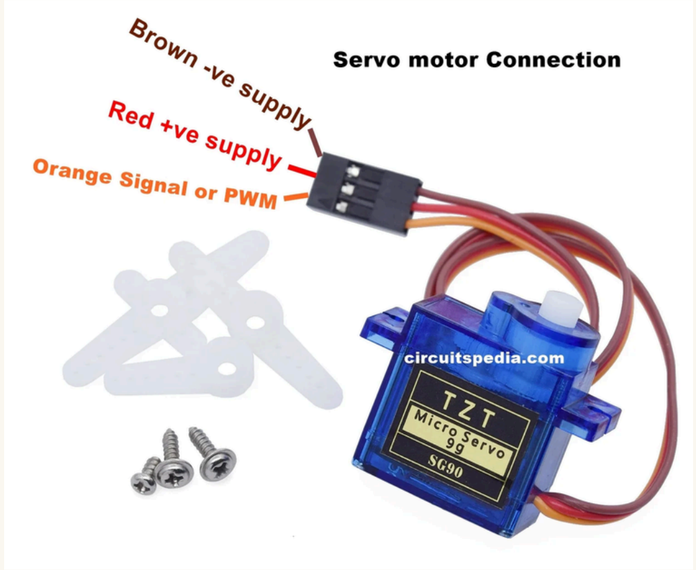
RTC Module-DS3231

- Operating voltage: 3.3V – 5V
- Communication: I2C protocol
- Accuracy: ±2 ppm (very high accuracy)
- Interface pins: SDA, SCL
- Backup: Built-in battery support
- Function: Provides real-time clock



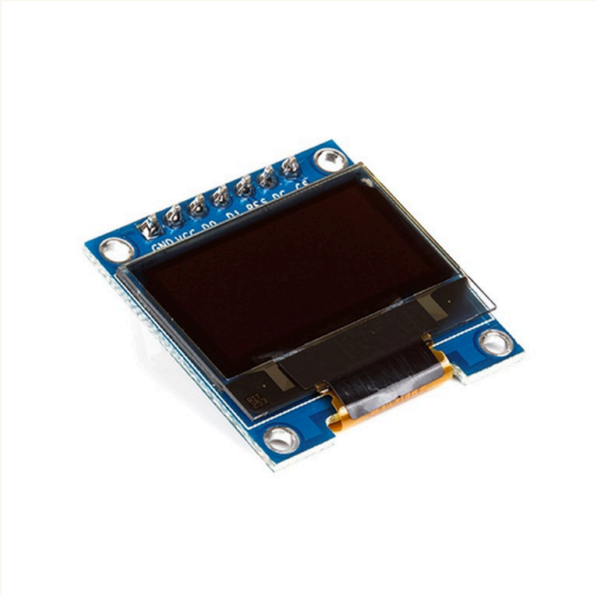
Servo Motor – SG90 Micro Servo

- Operating voltage: 4.8V – 6V
- Rotation angle: 0° to 180°
- Control: PWM signal
- Torque: ~1.8 kg·cm
- Speed: 0.1 sec/60°
- Function: Controls gate opening



OLED Display (0.96” I2C OLED – SSD1306)

- Display type: OLED (Organic Light Emitting Diode)
- Size: 0.96 inch
- Resolution: 128 × 64 pixels
- Operating voltage: 3.3V – 5V
- Communication: I2C protocol
- Interface pins: VCC, GND, SDA, SCL
- Controller: SSD1306
- Power consumption: Low (~20 mA)
- Contrast: High (self-emitting pixels, no backlight)



SOFTWARE SPECIFICATIONS

TELEGRAM BOT

A cloud-based messaging service that enables real-time notifications and interaction with users through the Telegram platform. It supports API-based communication with ESP32 over Wi-Fi to send alerts and receive user confirmations remotely

Arduino IDE 2.3.8

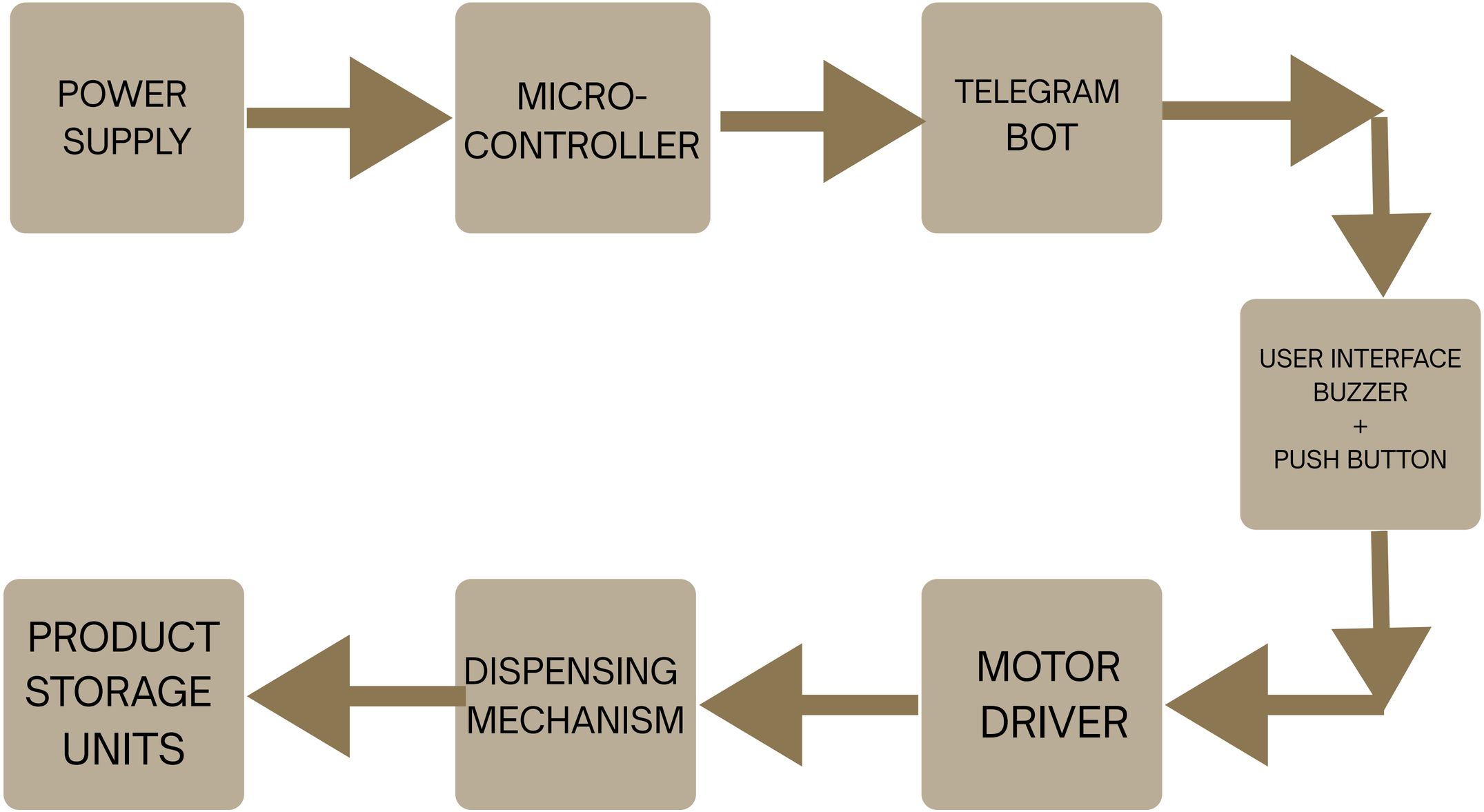
An open-source development environment used for writing, compiling, and uploading code to the ESP32 microcontroller. It provides built-in library support, serial monitoring, and an easy interface for embedded system development.

SOFTWARE SPECIFICATIONS

C

A high-performance programming language used to implement the control logic of the system. It enables efficient handling of hardware operations, timing control, and integration with external modules like sensors and communication interfaces.

BLOCK DIAGRAM



DESIGN

Dispenser Size

- Width: 200 mm
- Depth: 40mm
- Height: 150 mm

Material used

- Foam board,cardboard

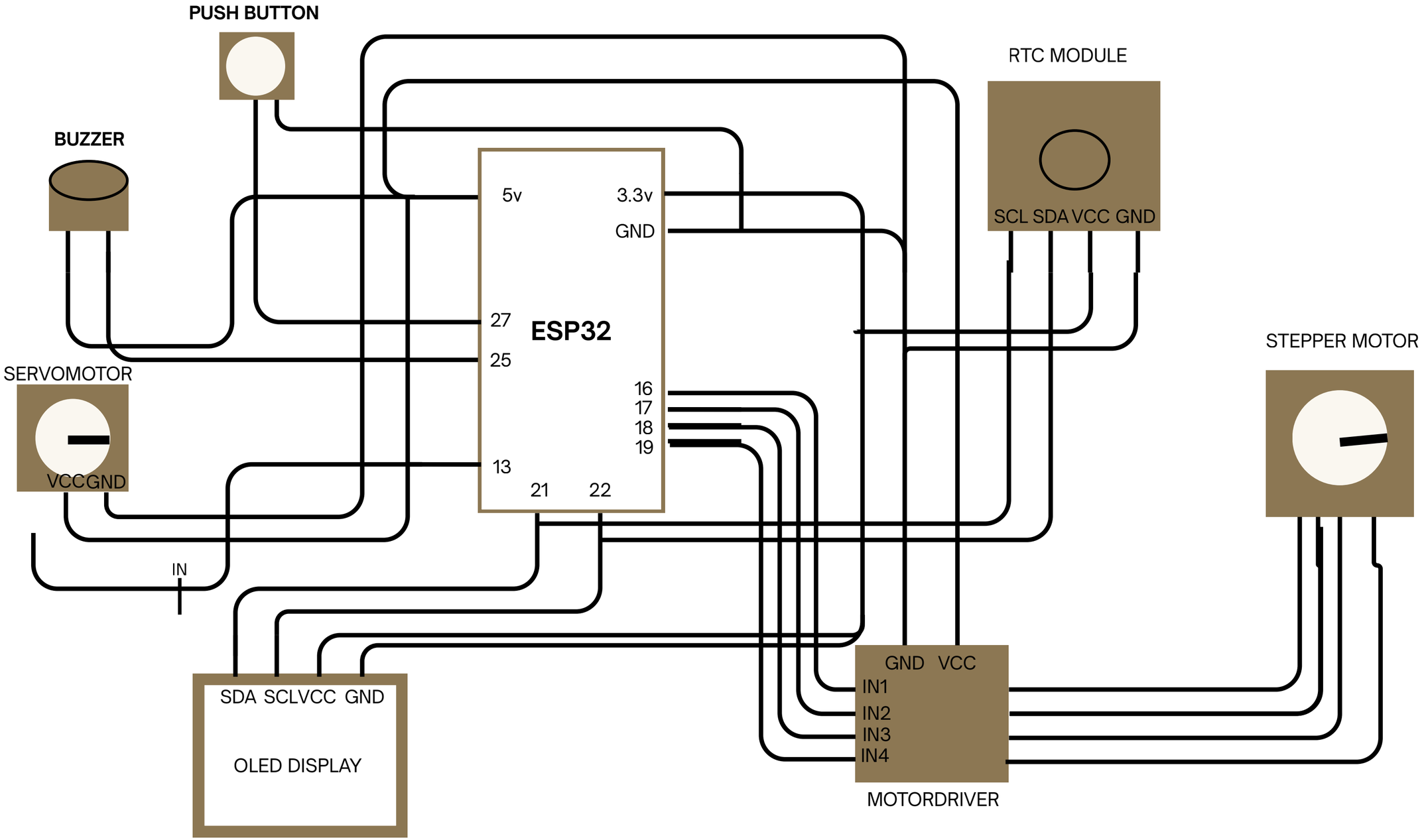
Individual Compartment Size

- Compartment Length (arc length): ~20 mm
- Compartment Width: 28mm
- Compartment Depth: 40mm
- No.of Compartments: 9(1 for exit hole)

DESIGN

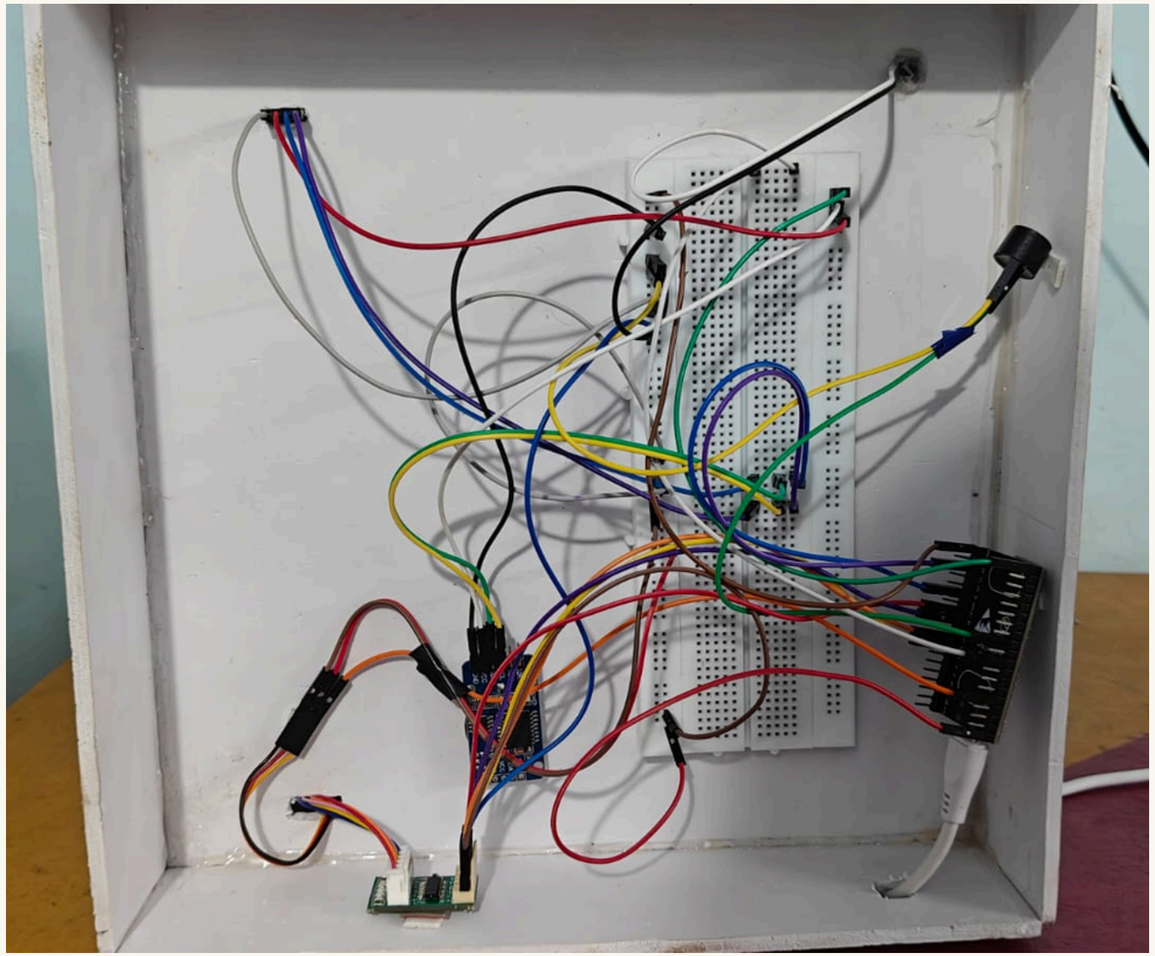
Dispensing Mechanism

- Stepper rotates compartment holder to align one compartment with outlet
- Servo actuates a trapdoor / sliding gate
- Gate opens once → all 3 pills drop together
- Gate closes before next index
- One dispense = exactly one compartment


PUSH BUTTON
BUZZER
RTC MODULE
SCL SDA VCC GND
5v
3.3v
GND
27
25
ESP32
16
17
18
19
13
21
22
SERVOMOTOR
VCCGND
IN
STEPPER MOTOR
SDA SCLVCC GND
OLED DISPLAY
GND VCC
IN1
IN2
IN3
IN4
MOTORDRIVER

PROTOTYPE IMAGES





HARDWARE CHALLENGE

- Precise alignment of the stepper motor to ensure correct pill compartment positioning.
- Synchronization between stepper motor and servo motor for accurate dispensing.
- Power supply stability when multiple components (motors, display, Wi-Fi) operate simultaneously.
- Mechanical wear and tear of moving parts like gears and dispensing mechanism.
- Noise and vibration generated by motors affecting system reliability.
- Ensuring compact design while integrating all hardware components efficiently.

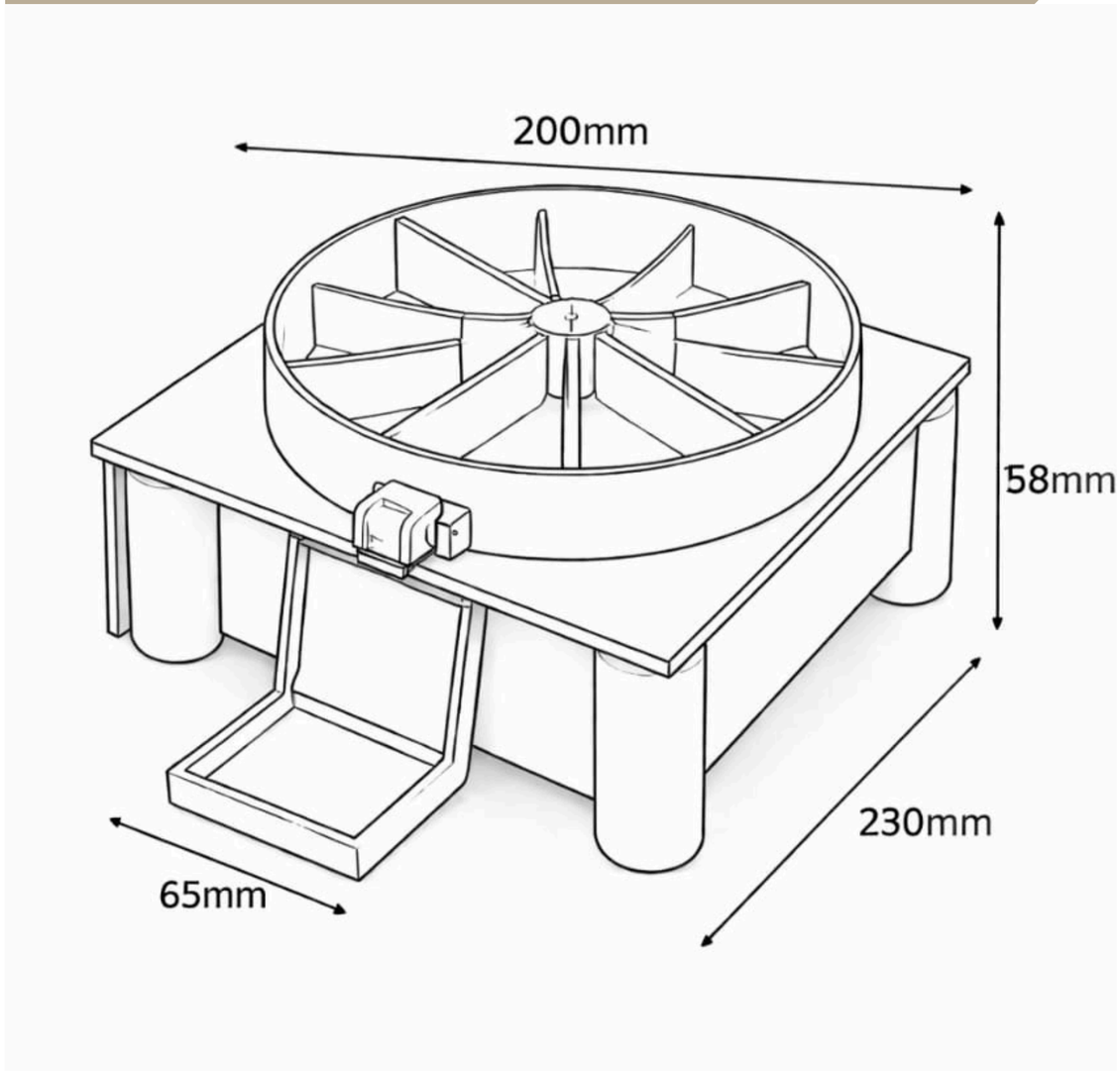
LIMITATIONS

- Limited storage capacity for pills depending on the mechanical design.
- Dependence on power supply and internet (for alerts), failure may affect functionality.

FUTURE SCOPE

- Integration of sensors (weight/IR) to confirm actual pill consumption automatically.
- Mobile app development for better user interface and control instead of only Telegram.
- Voice assistant integration for elderly-friendly interaction.
- AI-based scheduling and dosage adjustment based on user health data.
- Battery backup system for uninterrupted operation during power failure.
- Compact and portable design with improved mechanical reliability.

LAYOUT



COST ESTIMATION

Component	Model Name	Estimated Cost (₹)
ESP32	ESP32 DevKit V1	₹530
Stepper Motor	28BYJ 48	₹150
Motor Driver	ULN2003 Driver Board	₹45
RTC Module	DS3231	₹120
OLED Display	SSD1306 (128x64)	₹190
Servo Motor	SG90	₹100
Buzzer	Active Buzzer Module	₹26
Push Button	Tactile Switch	₹2
IR Sensor	IR Obstacle Sensor	₹40
Acrylic / Plywood	3mm thick sheets	₹200
Breadboard	Standard Solderless	₹100
Adapter	5V DC Power Supply	₹149
Adapter	5V DC Power Supply	₹149
Total Estimated Cost		**₹1,652**

NB:The prices mentioned above are rough estimates for single units and are subject to change.

COMPONENTS

Component	Model Name	Operating Voltage
ESP32	ESP32 DevKit V1	3.3V – 5V
Stepper Motor	28BYJ-48	5V
Motor Driver	ULN2003 Driver Board	5V
RTC Module	DS3231	3.3V – 5V
OLED Display	SSD1306 (128×64)	3.3V – 5V
Buzzer	Active Buzzer Module	3V – 5V
Push Button	Tactile Switch	3.3V – 5V
Acrylic / Plywood	3mm thick sheets	–
Breadboard	Standard Solderless	–
Adapter	5V DC Power Supply	5V
Servo Motor	SG90 Micro Servo	4.8V – 6V (5V typical)

POWER SUPPLY REQUIREMENTS

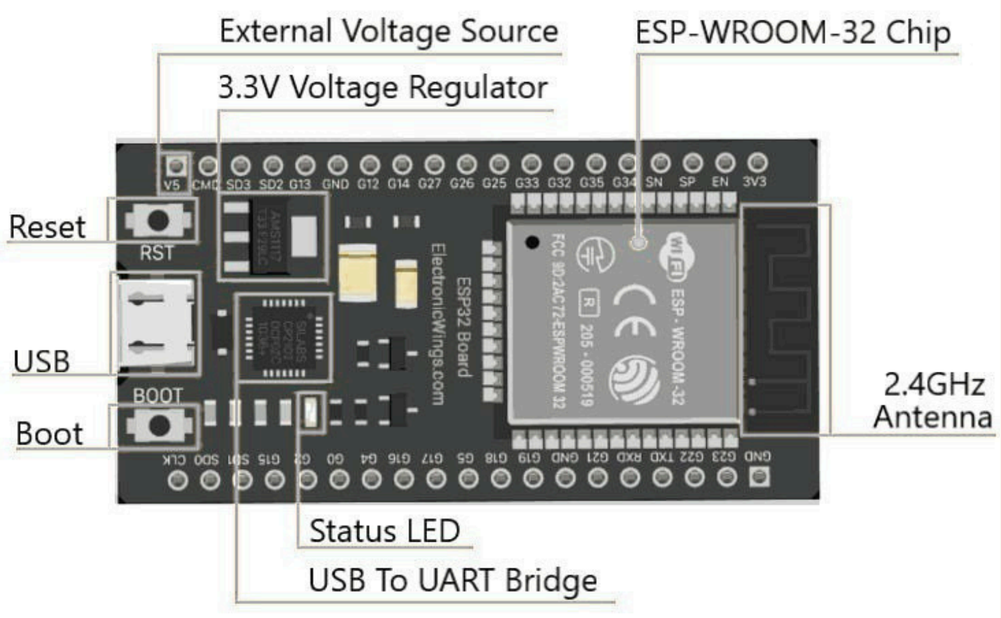
COMPONENTS	OPERATING VOLTAGE(V)	MAXIMUM CURRENT (I)	POWER(P=V*I)
ESP32	5	500mA	2.5W
28BYJ-48 Stepper & SG90 Servo	5	240mA	2.4W
ULN2003	5	240mA	1.2W
Buzzer	5	40mA	0.20W
OLED	3.3	20mA	0.07W

TOTAL	**1.04A**	**6.37W**

- Total Power Required = **6.53 W**
- With Engineering Safety Margin

  6.53×1.8=11.75W
- The Power supply required is **5V 2A**

MICROCONTROLLER



- ESP32 Development Board
- Model: ESP32 Dev Module
- Operating Voltage: 3.3V
- Wi-Fi: 802.11 b/g/n
- Clock Speed: Up to 240 MHz

Why ESP32?

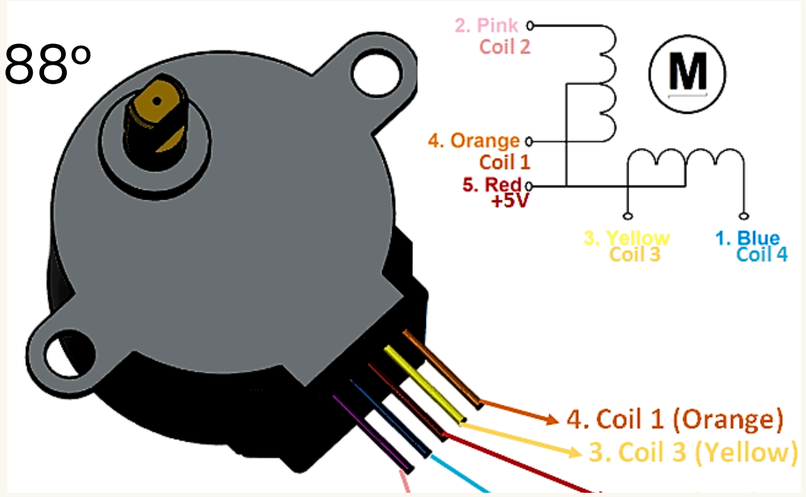
*Built-in WiFi

*Faster than Arduino

*Perfect for IoT

STEPPER MOTOR

- Model: 28BYJ-48
- Voltage: 5 V
- Step Angle: 5.625° / 64 (≈ 0.088° per step)
- Rotation: 360° (full rotation)



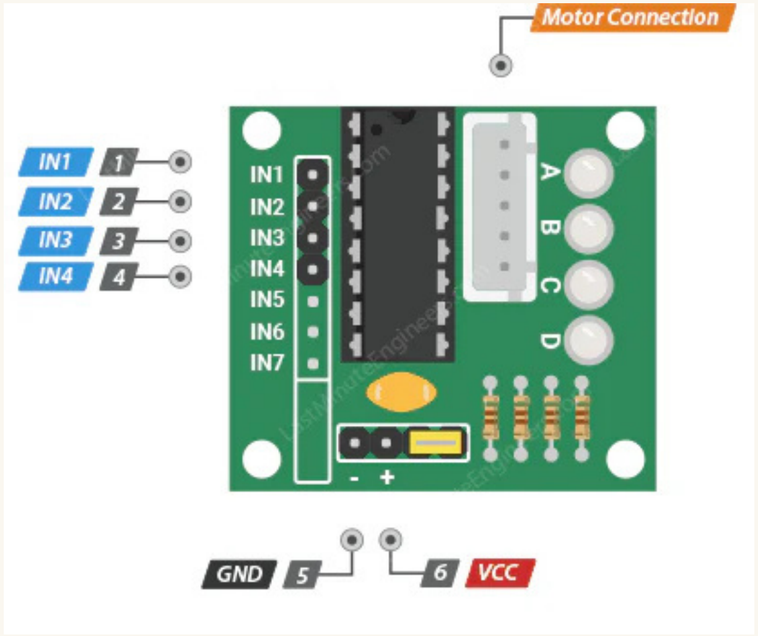


Why best?

1. High positional accuracy
2. No feedback required
3. Precise pill dispensing control
4. Smooth and repeatable motion

MOTOR DRIVER





- Model: ULN2003
- Voltage: 5 V
- Channel Count: 7-channel Darlington array
- Compatible Motor: 28BYJ-48 Stepper Motor

<u>Why best?</u>

1. Directly compatible with 28BYJ-48
2. Handles higher current safely
3. Simple interfacing with ESP32
4. Built-in flyback diodes for protection

SERVO MOTOR

- Model: SG90
- Voltage: 5 V
- Rotation: 0° – 180°
- Torque: ~1.8 kg·cm

Why best?

1. Precise angle control
2. Simple PWM control
3. Easy to interface with ESP32
4. Low cost and compact size



CONCLUSION

In conclusion, our electronic medicine dispenser is here to make taking medication easier for everyone. It helps people remember when to take their pills. On the hardware side, the peripheral devices such as servo motor has effectively done their function. We think that if people use our new medicine dispenser every day, they'll be better at taking their medicine like the doctor says. This means they'll get healthier and feel better overall. So, it's like having a helpful friend reminding you to take care of yourself.

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THANK YOU